Lab report - Microscopy Imaging - How to build a virtual microscope

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1 Part 3

Q.4 In brightfield, the unscattered light dominates and the transparent cells are barely visible (only the absorbing nuclei and a faint edge). In darkfield, the annular pupil blocks the unscattered (zero-frequency) light: the background becomes black and only light scattered by the cells reaches the camera, so the cells appear bright, mostly at their edges where the phase gradient (light deflection) is strongest. Darkfield reveals the cell outlines but does not give a quantitative phase measurement.